QF620 Stochastic Modelling in Finance — Session 4 Reference Cheatsheet

Stochastic Integrals and Itô's Formula: building stochastic models, SDEs, Riemann vs stochastic integrals, previsibility, Itô isometry, box calculus, Itô's formula

Built from Session 4 slides (Tee Chyng Wen, SMU) · October 2026 · revised edition with intuition and full slide derivations

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Source: "Session 4: Stochastic Integrals and Itô's Formula" (Tee Chyng Wen, SMU, 23 slides).

How to read this sheet. Each concept entry has up to five parts: **The idea in words** (what the concept is and why it is there), the formula with a shaded where box naming every symbol, **Derivation** (the slide's argument line by line, with the reason for each step in brown), **Worked example**, and **Exam angle** (traps

and what to be able to reproduce). Slide numbers are PDF page numbers, shown as [S#]. Material not on the slides is tagged *[Beyond slides]*. △! marks a slide ambiguity or error, with the corrected version.

What you must be able to do after this session: read an SDE as an integral equation; explain why the stochastic integral uses left endpoints; prove $\mathbb{E}[\int f\,dW] = 0$ and the Itô isometry from the definition; compute means and variances of stochastic integrals; use the box-calculus table; derive Itô's formula from a second-order Taylor expansion; apply both versions of Itô's formula ($f(X_t)$ and $g(t, X_t)$).

Notation used throughout (slide conventions)

Symbol	Meaning	Notes
W_t	Brownian motion	Session 3
S_t	stock price process	[S2]
$\mu,\ \mu(t,S_t),\ \mu_t$	drift (constant / state-dependent / general)	[S2, S3, S22]
$\sigma,\ \sigma(t,S_t),\ \sigma_t$	volatility (diffusion coefficient)	[S2, S3, S22]
$dt,\ dW_t$	infinitesimal time step; Brownian increment	[S5]
$\Delta t,\ \Delta W_t$	finite step; $W_{t+\Delta t} - W_t \sim \mathcal{N}(0, \Delta t)$	[S4]
P_n	partition $0 = t_0 < t_1 < \cdots < t_n = T$	[S6]
I_T	stochastic integral $\int_0^T f(u, W_u)\,dW_u$	[S9]
f', f''	derivatives in the state variable	[S19–S22]
g_t, g_x, g_{xx}	partial derivatives of $g(t,x)$	[S23]

0. Topic map

```
PREREQUISITES: S3 Brownian motion (independent NΔ(0,t) increments, nowhere differentiable)
    ▼
SESSION 4├─
 MODELS [–S2S5]│
     ├─ S_t = ₀S + μt + σW_t ~ N(₀Sμ+t, σ²t)                                  [S2]│
     ├─ integral form ⇔ differential form dS = μ dt + σ dW                     [S3]│
     └─ deterministic + stochastic parts ; dS/dt does not exist                [–S4S5]├─
 STOCHASTIC INTEGRAL [–S6S12]│
     ├─ Riemann: Σ f(x_i)Δt_i, any x_i                                         [–S6S7]│
     ├─ Itô: Σ f(t_{–i1})ΔW_i, LEFT endpoint (previsibility)                   [S8]│
     ├─ E∫[f dW] = 0 ; E∫[(f dW)²] = E∫[f² dt] (isometry) ; proofs             [–S9S11]│
     └─ example ∫W dW: mean 0, variance T²/2                                   [S12]├─
 BOX CALCULUS [–S13S14]│
     └─ law of large numbers ⇒ Δ(W)² ≈ Δt ; dt² = 0, dt dW = 0, dW² = dt       [–S13S14]└─
 ITÔ'S FORMULA [–S15S23]
      ├─ history; why we need it                                               [–S15S18]
      ├─ ordinary Taylor dx = ′fdt ; stochastic dX = ′fdW + ″½fdt              [–S19S20]
      ├─ dX = μ(′f + σ″½²f)dt + σ′fdW                                           [–S21S22]
      └─ with time: dY = (g_t + μg_x + σ½²g_xx)dt + σg_x dW                     [S23]
    ▼
LATER: S5 solving SDEs (GBM via d ln S) ; S6 –BlackScholes PDE ; S7 Girsanov
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The storyline of the session.

- We want a stock model with controllable drift and volatility: $dS = \mu\,dt + \sigma\,dW$. Since Brownian paths have no derivative, this is shorthand for an **integral** equation, and the $\int \sigma\,dW$ part needs a new kind of integral.
- The **stochastic integral** is a Riemann-like sum but evaluated at the *left* endpoint (you choose your position before the price moves). Independent, mean-zero increments then give its two key properties: mean zero and the Itô isometry.
- $(\Delta W)^2$ behaves like Δt, not like $(\Delta t)^2$, so in a Taylor expansion the second-order term survives. That extra $\frac{1}{2}f''\,dt$ is **Itô's formula**, the source of every $-\frac{1}{2}\sigma^2$ in the course.

1 Concept entries

1.1 Building stochastic models [S2]

The idea in words
Plain W_t is a poor stock model: it has no trend and its randomness scale is fixed at 1. Add a deterministic trend μt (drift) and scale the noise by σ (volatility). The result is still normal: a shifted, stretched Brownian motion.

$$S_t = S_0 + \mu t + \sigma W_t, \quad \mu, \sigma \in \mathbb{R}; \qquad S_t \sim \mathcal{N}(S_0 + \mu t,\ \sigma^2 t), \quad \mathbb{E}[S_t] = S_0 + \mu t, \quad V[S_t] = V[\sigma W_t] = \sigma^2 t.$$

where

μt	drift: deterministic trend, μ per unit time.
σW_t	random part: Brownian motion scaled by the volatility.
$\mathbb{E}[S_t]$	$\mathbb{E}[W_t] = 0$, so only the deterministic part survives.
$V[S_t]$	constants add no variance; $V[\sigma W_t] = \sigma^2 V[W_t] = \sigma^2 t$.

[Beyond slides] Weakness: normally distributed, so S_t can go negative: $\mathbb{P}(S_t < 0) = \Phi\big(-\frac{S_0+\mu t}{\sigma\sqrt{t}}\big) > 0$ (Bachelier model).

1.2 Integral and differential forms; SDEs [S3]

The idea in words
"$\mathrm{d}S_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$" is a short way of writing "the total change of S from 0 to t is the accumulated drift plus the accumulated random shocks". In real models the drift and volatility can depend on the current price and time, e.g. GBM has $\mu(t,S) = \mu S$, $\sigma(t,S) = \sigma S$.

$$S_t = S_0 + \int_0^t \sigma\,\mathrm{d}W_u + \int_0^t \mu\,\mathrm{d}u \quad \Longleftrightarrow \quad \mathrm{d}S_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t; \qquad \text{in general} \quad \mathrm{d}S_t = \mu(t,S_t)\,\mathrm{d}t + \sigma(t,S_t)\,\mathrm{d}W_t.$$

where

$\int_0^t \mu\,\mathrm{d}u$	accumulated drift (ordinary integral) $= \mu t$.
$\int_0^t \sigma\,\mathrm{d}W_u$	accumulated random shocks (stochastic integral) $= \sigma W_t$ for constant σ.
$\mu(t,S_t),\ \sigma(t,S_t)$	drift and volatility coefficients: functions of time and the current price only.

With constant μ, σ the SDE is solved by $S_t = S_0 + \mu t + \sigma W_t$.

Exam angle
△! [S3] says "a stochastic process is a continuous process that can be written" in this form. That describes an *Itô process* (diffusion); many processes (e.g. with jumps) are not of this form.

1.3 Deterministic and stochastic components [S4–S5]

The idea in words
Over a small step, the future price is "present + drift × time + volatility × Brownian increment". The drift uses the price at the *start* of the step, which is already known: that is why it counts as deterministic over the step. Only the increment is unknown, and it is independent of everything so far. We cannot divide by $\mathrm{d}t$ to get an ODE, because $\mathrm{d}W/\mathrm{d}t$ does not exist.

$$\underbrace{S_{t+\Delta t}}_{\text{future}} = \underbrace{S_t}_{\text{present}} + \underbrace{\mu(t,S_t)\Delta t}_{\text{deterministic}} + \underbrace{\sigma(t,S_t)(W_{t+\Delta t} - W_t)}_{\text{stochastic}} \xrightarrow{\Delta t \to 0^+} \mathrm{d}S_t = \mu(t,S_t)\,\mathrm{d}t + \sigma(t,S_t)\,\mathrm{d}W_t.$$

where

$W_{t+\Delta t} - W_t$	Brownian increment $\sim \mathcal{N}(0, \Delta t)$, independent of everything up to t.
$\frac{\mathrm{d}W_t}{\mathrm{d}t}$	does not exist: W is nowhere differentiable, so the SDE must be read as an integral equation [S5].

Slide question [S4] (answered here): $\mu(t, S_t)$ is evaluated at the start of the step, where S_t is known. Given information at t, the drift over $[t, t + \Delta t]$ is a known number; only the Brownian increment is random.

1.4 Integral form and the Riemann integral [S6–S7]

The idea in words

The integral form has two pieces. The $\mathrm{d}u$ integral is the familiar area under a curve (Riemann sum of rectangles); for a nice function it does not matter where in each sub-interval you evaluate the height. The $\mathrm{d}W$ integral looks similar but the "widths" are random Brownian increments, and there the evaluation point *does* matter.

$$S_T = S_0 + \underbrace{\int_0^T \mu(u, S_u)\,\mathrm{d}u}_{\text{Riemann integral}} + \underbrace{\int_0^T \sigma(u, S_u)\,\mathrm{d}W_u}_{\text{stochastic integral}}; \qquad \int_0^T f\,\mathrm{d}t = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i)(t_i - t_{i-1}),\ x_i \in [t_{i-1}, t_i].$$

where

$t_i - t_{i-1}$	width of the i-th sub-interval of the partition P_n.
x_i	any point inside it (left, right, middle: same limit for a regular f).
$f(x_i)(t_i - t_{i-1})$	area of the i-th rectangle [S7].

1.5 The stochastic integral and previsibility [S8]

The idea in words

Replace the width $t_i - t_{i-1}$ by the Brownian increment $W_{t_i} - W_{t_{i-1}}$ and evaluate the integrand at the **left endpoint**. Financially: f is your position, ΔW is the price move; you must choose your position *before* the move happens (previsibility), so the integral is your trading P&L. Because W is so rough, other evaluation points give a different limit (e.g. midpoint gives the Stratonovich integral), so the choice is part of the definition. The result is a random variable, not a number.

$$\int_0^T f(t)\,\mathrm{d}W_t = \lim_{n\to\infty} \sum_{i=1}^{n} f(t_{i-1}) \times \left(W_{t_i} - W_{t_{i-1}}\right).$$

where

$f(t_{i-1})$	integrand at the left endpoint of each interval.
$W_{t_i} - W_{t_{i-1}}$	Brownian increment, playing the role of the width.
lim	exists "in an appropriate sense" (mean square, L^2 *[Beyond slides]*), not path by path.

Why the evaluation point matters *[Beyond slides]*

For $f = W$: left-point sum $\sum W_{t_{i-1}} \Delta W_i$ has mean 0. Right-point sum $\sum W_{t_i} \Delta W_i = \sum W_{t_{i-1}} \Delta W_i + \sum (\Delta W_i)^2$ has mean $0 + \sum \Delta t_i = T$. The difference $\sum (\Delta W_i)^2 \to T$ does not vanish, unlike in ordinary calculus where $\sum (\Delta t_i)^2 \to 0$.

1.6 Properties of stochastic integrals [S9]

The idea in words

Each term of the sum is "something known" × "a fresh mean-zero increment". So the integral has mean zero (a fair game: you cannot make money on average trading a martingale). For the second moment, cross terms

between different increments vanish and each diagonal term contributes $f^2 \times \Delta t$, because $\mathbb{E}[(\Delta W)^2] = \Delta t$. Hence "$\mathrm{d}W$ squared becomes $\mathrm{d}t$" inside expectations: the **Itô isometry**. It is the workhorse for variances of stochastic integrals.

For $I_T = \int_0^T f(u, W_u)\,\mathrm{d}W_u$:

$$
\begin{aligned}
&(1)\quad \mathbb{E}[I_T] = 0\\
&(2)\quad \mathbb{E}[I_T^2] = \mathbb{E}\Big[\Big(\int_0^T f(u,W_u)\,\mathrm{d}W_u\Big)^2\Big] = \mathbb{E}\Big[\int_0^T f(u,W_u)^2\,\mathrm{d}u\Big]\\
&(3)\quad f \text{ deterministic } \Rightarrow\ I_T \sim \mathcal{N}\Big(0, \int_0^T f(u)^2\,\mathrm{d}u\Big)\\
&(4)\quad \mathbb{E}\Big[\int_0^T f(u)\,\mathrm{d}W_u \times \int_0^T g(s)\,\mathrm{d}W_s\Big] = \mathbb{E}\Big[\int_0^T f(u)g(u)\,\mathrm{d}u\Big]\\
&(5)\quad \text{Itô isometry: } \mathbb{E}\Big[\Big(\int_0^T X_t\,\mathrm{d}W_t\Big)^2\Big] = \mathbb{E}\Big[\int_0^T X_t^2\,\mathrm{d}t\Big]
\end{aligned}
$$

where

$f(u, W_u), X_t$	integrand; may depend on time and the path so far (adapted / previsible).
$\mathbb{E}[I_T^2]$	since the mean is 0, this is the variance of I_T.
(3)	deterministic integrand: a sum of independent normals, hence normal.
(4)	covariance of two integrals against the same W; $g = f$ recovers (2).

Exam angle

△! [S9] Property (3) needs a deterministic f (with a random integrand I_T is usually not normal, e.g. $\int W\,\mathrm{d}W$). (2) and (5) are the same statement. Typical use: $\int_0^T t\,\mathrm{d}W_t \sim \mathcal{N}(0, \int_0^T t^2\,\mathrm{d}t) = \mathcal{N}(0, T^3/3)$.

1.7 Proofs of the mean and the isometry [S10–S11]

Proof of (1): $\mathbb{E}[\int f\,\mathrm{d}W] = 0$ [S10]

$$
\begin{aligned}
\mathbb{E}\Big[\int_0^T f(t,W_t)\,\mathrm{d}W_t\Big] &= \mathbb{E}\Big[\lim_{n\to\infty}\sum_{i=1}^n f(t_{i-1}, W_{t_{i-1}})(W_{t_i} - W_{t_{i-1}})\Big] && \text{(definition)}\\
&= \lim_{n\to\infty}\sum_{i=1}^n \mathbb{E}\big[f(t_{i-1}, W_{t_{i-1}})(W_{t_i} - W_{t_{i-1}})\big] && \text{(swap } \mathbb{E}\text{, lim, } \textstyle\sum\text{)}\\
&= \lim_{n\to\infty}\sum_{i=1}^n \mathbb{E}\big[f(t_{i-1}, W_{t_{i-1}})\big]\,\mathbb{E}\big[W_{t_i} - W_{t_{i-1}}\big] && \text{(increment indep. of } W_{t_{i-1}}\text{)}\\
&= \lim_{n\to\infty}\sum_{i=1}^n \mathbb{E}\big[f(t_{i-1}, W_{t_{i-1}})\big] \times 0 = 0 && \text{(mean-zero increments)}
\end{aligned}
$$

This is exactly where the left endpoint is used: $f(t_{i-1}, \cdot)$ depends only on the path up to t_{i-1}.

Proof of (2), the isometry [S11]

Write $\Delta W_i = W_{t_i} - W_{t_{i-1}}$ and $f_i = f(t_{i-1}, W_{t_{i-1}})$. Key lemma:

$$
\mathbb{E}[f_i f_j\,\Delta W_i\,\Delta W_j] = \begin{cases} \mathbb{E}[f_i^2]\,(t_i - t_{i-1}) & i = j \\ 0 & i \neq j \end{cases}
$$

Case $i \neq j$, say $i > j$: ΔW_i is independent of $f_i, f_j, \Delta W_j$ (all known by t_{i-1}), so the expectation factorises with $\mathbb{E}[\Delta W_i] = 0$. *Case* $i = j$: ΔW_i independent of f_i, and $\mathbb{E}[(\Delta W_i)^2] = t_i - t_{i-1}$. Then

$$\mathbb{E}\Big[\Big(\int_0^T f\,\mathrm{d}W\Big)^2\Big] = \mathbb{E}\Big[\int_0^T f(s,W_s)\,\mathrm{d}W_s \times \int_0^T f(u,W_u)\,\mathrm{d}W_u\Big] \qquad \text{(square = product)}$$

$$= \lim_{n\to\infty}\sum_{i=1}^{n}\sum_{j=1}^{n}\mathbb{E}[f_i f_j\,\Delta W_i \Delta W_j] \qquad \text{(multiply out the sums)}$$

$$= \lim_{n\to\infty}\sum_{i=1}^{n}\mathbb{E}[f_i^2]\,(t_i - t_{i-1}) \qquad \text{(only diagonal survives)}$$

$$= \mathbb{E}\Big[\int_0^T f(t,W_t)^2\,\mathrm{d}t\Big] \qquad \text{(a Riemann sum)}$$

Exam angle
△! [S10–S11] swap $\mathbb{E}$ and lim without comment; this is justified by mean-square convergence of the sums *[Beyond slides]*. In the exam, reproduce the two cases ($i=j$, $i\neq j$) with the independence reason.

1.8 Example: $\int_0^T W_t\,\mathrm{d}W_t$ [S12]

Mean and variance (worked here; the slide only poses it)

$$\mathbb{E}[I_T] = 0 \qquad \text{(property (1))}$$

$$V[I_T] = \mathbb{E}[I_T^2] = \mathbb{E}\Big[\int_0^T W_t^2\,\mathrm{d}t\Big] \qquad \text{(isometry with } f = W_t\text{)}$$

$$= \int_0^T \mathbb{E}[W_t^2]\,\mathrm{d}t = \int_0^T t\,\mathrm{d}t = \frac{T^2}{2} \qquad \text{(swap } \mathbb{E} \text{ and } \textstyle\int \mathrm{d}t\text{; } \mathbb{E}[W_t^2] = t\text{)}$$

[Beyond slides] Closed form via Itô on W_t^2 (1.12): $\int_0^T W\,\mathrm{d}W = \frac{1}{2}W_T^2 - \frac{1}{2}T$. Check: mean $\frac{1}{2}T - \frac{1}{2}T = 0$; variance $\frac{1}{4}V[W_T^2] = \frac{1}{4}\cdot 2T^2 = \frac{T^2}{2}$ ✓. Ordinary calculus would wrongly give $\frac{1}{2}W_T^2$ (mean $T/2 \neq 0$).

1.9 Box calculus [S13–S14]

The idea in words
Over a step of length Δt, $(\Delta W)^2$ is random but its mean is exactly Δt; summed over many steps, the fluctuations average out (law of large numbers) and $\sum(\Delta W)^2 \to T$ *deterministically*. So in the limit, $(\mathrm{d}W)^2$ can be replaced by $\mathrm{d}t$. Anything of higher order than $\mathrm{d}t$ ($\mathrm{d}t^2$, $\mathrm{d}t\,\mathrm{d}W \sim \mathrm{d}t^{3/2}$) is negligible. This multiplication table is all you need to compute $(\mathrm{d}X)^2$ in Itô's formula.

Strong law of large numbers [S13]: $\mathbb{P}\Big(\lim_{n\to\infty}\frac{X_1+\cdots+X_n}{n} = \mu\Big) = 1$; e.g. $X_i = \pm 1$ equally likely $\Rightarrow \frac{X_1+\cdots+X_n}{n} \to 0$.

Why $(\Delta W)^2 \approx \Delta t$ *[Beyond slides]* (the slide's argument made explicit)

$$\Delta W_i = \sqrt{\Delta t}\,Z_i,\ Z_i \text{ i.i.d. } \mathcal{N}(0,1), \quad \Delta t = T/n \qquad \text{(BM increments)}$$

$$\textstyle\sum_{i=1}^n (\Delta W_i)^2 = \Delta t \sum Z_i^2 = T\cdot\frac{1}{n}\sum_{i=1}^n Z_i^2 \ \to\ T\cdot\mathbb{E}[Z^2] = T \qquad \text{(strong law)}$$

$$\mathbb{E}[(\Delta W)^2] = \Delta t, \qquad V[(\Delta W)^2] = \Delta t^2\,V[Z^2] = 2\Delta t^2 \qquad \text{(fluctuation is } O(\Delta t^2)\text{)}$$

So $(\Delta W)^2$ is of order Δt (its randomness is an order smaller and washes out), and ΔW is of order $\sqrt{\Delta t}$. The limit $\sum(\Delta W)^2 \to T$ is the *quadratic variation* of W.

Rules [S14]: $(\mathrm{d}t)^2 = 0$, $\mathrm{d}t\cdot\mathrm{d}W_t = 0$, $(\mathrm{d}W_t)^2 = \mathrm{d}t$.

$\cdot$	$\mathrm{d}t$	$\mathrm{d}W_t$
$\mathrm{d}t$	0	0
$\mathrm{d}W_t$	0	$\mathrm{d}t$

Example [S14]: $(\mathrm{d}X)^2$
$\mathrm{d}X_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t \;\Rightarrow\; (\mathrm{d}X_t)^2 = \mu^2(\mathrm{d}t)^2 + 2\mu\sigma\,\mathrm{d}t\,\mathrm{d}W_t + \sigma^2(\mathrm{d}W_t)^2 = 0 + 0 + \sigma^2\,\mathrm{d}t$. Only the volatility matters. *[Beyond slides]*: for GBM $(\mathrm{d}S)^2 = \sigma^2 S^2\,\mathrm{d}t$; for correlated BMs $\mathrm{d}W^1\,\mathrm{d}W^2 = \rho\,\mathrm{d}t$.

1.10 History and motivation [S15–S18]

- Brown (1827) pollen; Einstein, Perrin (early 1900s); Wiener (1923) probability measures on path space; Kiyosi Itô (1942) built the stochastic integral and SDEs from scratch [S15–S16].
- **Why Itô's formula [S17]:** theoretically, to manipulate stochastic integrals with chain/product rules instead of going back to the definition; practically, an option price is a *function* of the stock price process, so we need the dynamics of a function of a stochastic process.
- Quotations [S18]: Hamming ("computing is insight, not numbers"), Föllmer (Itô calculus transformed quantitative finance).

1.11 Deriving Itô's formula [S19–S21]

The idea in words
The chain rule comes from a Taylor expansion: keep the first-order term, drop the rest because $(\Delta t)^2$ is negligible. Do the same with a Brownian argument and the second-order term $(\Delta W)^2$ is *not* negligible: it is $\mathrm{d}t$ by box calculus. So the stochastic chain rule has one extra term, $\frac{1}{2}f''\,\mathrm{d}t$. Intuition: for convex f, random wiggles push $f(W)$ up on average (Jensen), and that average push is the drift $\frac{1}{2}f''$.

Ordinary calculus [S19]

$$
\begin{aligned}
f(t+\Delta t) &= f(t) + f'(t)\Delta t + \tfrac{1}{2!}f''(t)(\Delta t)^2 + \tfrac{1}{3!}f'''(t)(\Delta t)^3 + \cdots && \text{(Taylor)}\\
\Delta x &= f'(t)\Delta t + \tfrac{1}{2!}f''(t)(\Delta t)^2 + \cdots && (x = f(t))\\
\frac{\mathrm{d}x}{\mathrm{d}t} &= f'(t) && (\text{divide by } \Delta t,\ \Delta t \to 0^+)
\end{aligned}
$$

Function of Brownian motion [S20]

$$
\begin{aligned}
f(W_t + \Delta W_t) &= f(W_t) + f'(W_t)\Delta W_t + \tfrac{1}{2!}f''(W_t)(\Delta W_t)^2 + \cdots && \text{(Taylor in } W)\\
\Delta X_t &= f'(W_t)\Delta W_t + \tfrac{1}{2}f''(W_t)(\Delta W_t)^2 + \cdots && (X_t = f(W_t))\\
\mathrm{d}X_t &= f'(W_t)\,\mathrm{d}W_t + \underbrace{\tfrac{1}{2}f''(W_t)\,\mathrm{d}t}_{\text{extra term}} && ((\mathrm{d}W)^2 = \mathrm{d}t,\ (\mathrm{d}W)^3 = 0)
\end{aligned}
$$

Function of a stock [S21]: $\Delta S_t = \mu\Delta t + \sigma\Delta W_t$, $X_t = f(S_t)$

$$
\begin{aligned}
\Delta X_t &= f'(S_t)\Delta S_t + \tfrac{1}{2!}f''(S_t)(\Delta S_t)^2 + \cdots && \text{(Taylor in } S)\\
(\Delta S_t)^2 &= \mu^2(\Delta t)^2 + 2\mu\sigma(\Delta t)(\Delta W_t) + \sigma^2(\Delta W_t)^2 \;\to\; \sigma^2\,\mathrm{d}t && \text{(box calculus)}\\
\mathrm{d}X_t &= f'(S_t)\,\mathrm{d}S_t + \tfrac{1}{2}f''(S_t)\sigma^2\,\mathrm{d}t && \text{(substitute)}\\
&= f'(S_t)(\mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t) + \tfrac{1}{2}\sigma^2 f''(S_t)\,\mathrm{d}t\\
&= \Big(\mu f'(S_t) + \frac{\sigma^2}{2}f''(S_t)\Big)\,\mathrm{d}t + \sigma f'(S_t)\,\mathrm{d}W_t && \text{(collect } \mathrm{d}t \text{ and } \mathrm{d}W)
\end{aligned}
$$

where

$f'\,\mathrm{d}S$	first-order (ordinary chain rule) term.
$\frac{1}{2}f''\sigma^2\,\mathrm{d}t$	second-order Itô term from $(\mathrm{d}S)^2 = \sigma^2\,\mathrm{d}t$.
$\mu f' + \frac{1}{2}\sigma^2 f''$	drift of X: chain-rule drift plus convexity correction.
$\sigma f'$	volatility of X: stock's volatility times the sensitivity f' (a "delta").

1.12 Itô's formula (function of a stochastic process) [S22]

Statement. If $\mathrm{d}X_t = \mu_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t$ and $f : \mathbb{R} \to \mathbb{R}$ with f, f', f'' continuous, then $Y_t = f(X_t)$ satisfies

$$\mathrm{d}Y_t = \Big(\mu_t f'(X_t) + \tfrac{1}{2}\sigma_t^2 f''(X_t)\Big)\,\mathrm{d}t + \sigma_t f'(X_t)\,\mathrm{d}W_t.$$

where

μ_t, σ_t	drift and volatility of X (may be random, but known at t).
$\frac{1}{2}\sigma_t^2 f''$	Itô correction from $(\mathrm{d}X)^2 = \sigma_t^2\,\mathrm{d}t$.

Recipe: (i) write f, f', f''; (ii) $\mathrm{d}Y = f'\,\mathrm{d}X + \frac{1}{2}f''(\mathrm{d}X)^2$; (iii) substitute $\mathrm{d}X$ and $(\mathrm{d}X)^2 = \sigma^2\,\mathrm{d}t$; (iv) collect $\mathrm{d}t$ and $\mathrm{d}W$ terms.

Slide example [S22] (worked here): $\mathrm{d}X_t = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t$, $Y_t = X_t^2$
$f(x) = x^2$, $f' = 2x$, $f'' = 2$:

$$\mathrm{d}Y_t = 2X_t\,\mathrm{d}X_t + \tfrac{1}{2}\cdot 2\,(\mathrm{d}X_t)^2 = 2X_t(\mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t) + \sigma^2\,\mathrm{d}t = (2\mu X_t + \sigma^2)\,\mathrm{d}t + 2\sigma X_t\,\mathrm{d}W_t.$$

[Beyond slides] With $\mu = 0$, $\sigma = 1$ ($X = W$): $\mathrm{d}(W_t^2) = \mathrm{d}t + 2W_t\,\mathrm{d}W_t$. Integrate: $W_T^2 = T + 2\int_0^T W\,\mathrm{d}W$, so $\int_0^T W\,\mathrm{d}W = \frac{1}{2}W_T^2 - \frac{1}{2}T$, and $W_t^2 - t = 2\int_0^t W\,\mathrm{d}W$ is a martingale (Session 3).

1.13 Itô's formula with time [S23]

The idea in words
Option prices depend on time as well as the stock: $V(t, S)$. Expand in both variables. The time direction behaves like ordinary calculus (only $g_t\,\mathrm{d}t$ survives); the state direction gets the Itô correction. All mixed and higher terms ($g_{tt}\,\mathrm{d}t^2$, $g_{tx}\,\mathrm{d}t\,\mathrm{d}X$) vanish by the box table. This version gives the Black–Scholes PDE in Session 6.

Statement. If $\mathrm{d}X_t = \mu_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}W_t$ and $g(t, x)$ with g_t, g_x, g_{xx} continuous, then $Y_t = g(t, X_t)$ satisfies

$$\mathrm{d}Y_t = \Big(g_t(t, X_t) + \mu_t g_x(t, X_t) + \tfrac{1}{2}\sigma_t^2 g_{xx}(t, X_t)\Big)\,\mathrm{d}t + \sigma_t g_x(t, X_t)\,\mathrm{d}W_t.$$

Where it comes from *[Beyond slides]* (two-variable Taylor + box calculus)

$$\begin{aligned}
\mathrm{d}Y &= g_t\,\mathrm{d}t + g_x\,\mathrm{d}X + \tfrac{1}{2}g_{tt}(\mathrm{d}t)^2 + g_{tx}\,\mathrm{d}t\,\mathrm{d}X + \tfrac{1}{2}g_{xx}(\mathrm{d}X)^2 + \cdots && \text{(Taylor in } (t, x)\text{)} \\
&= g_t\,\mathrm{d}t + g_x(\mu_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}W) + 0 + 0 + \tfrac{1}{2}g_{xx}\sigma_t^2\,\mathrm{d}t && ((\mathrm{d}t)^2 = \mathrm{d}t\,\mathrm{d}X = 0) \\
&= \big(g_t + \mu_t g_x + \tfrac{1}{2}\sigma_t^2 g_{xx}\big)\,\mathrm{d}t + \sigma_t g_x\,\mathrm{d}W && \text{(collect)}
\end{aligned}$$

where

g_t	sensitivity to the passage of time ("theta").
g_x	sensitivity to the state ("delta").
g_{xx}	curvature in the state ("gamma"); produces the Itô term.

Slide example [S23] (worked here): $Y_t = e^{X_t + t}$
$g = e^{x+t}$, so $g_t = g_x = g_{xx} = e^{x+t} = Y$:

$$\mathrm{d}Y_t = \Big(1 + \mu + \tfrac{1}{2}\sigma^2\Big)Y_t\,\mathrm{d}t + \sigma Y_t\,\mathrm{d}W_t.$$

The 1 comes from g_t (explicit $+t$), μ from μg_x, $\frac{1}{2}\sigma^2$ from the Itô term. Volatility $\propto Y$: Y is a geometric Brownian motion.

GBM via Itô *[Beyond slides]* (preview of Session 5)
$\mathrm{d}S_t = \mu S_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t$, $f = \ln S$, $f' = 1/S$, $f'' = -1/S^2$:

$$\begin{aligned}\mathrm{d}(\ln S_t) &= \frac{1}{S_t}\,\mathrm{d}S_t - \frac{1}{2}\frac{1}{S_t^2}(\,\mathrm{d}S_t)^2 = (\mu\,\mathrm{d}t + \sigma\,\mathrm{d}W_t) - \tfrac{1}{2}\sigma^2\,\mathrm{d}t \\ &= \left(\mu - \tfrac{1}{2}\sigma^2\right)\mathrm{d}t + \sigma\,\mathrm{d}W_t \;\Rightarrow\; S_t = S_0 \exp\left[\left(\mu - \tfrac{1}{2}\sigma^2\right)t + \sigma W_t\right].\end{aligned}$$

This is the $r - \frac{1}{2}\sigma^2$ of Session 2 when $\mu = r$.

1.14 Implementation *[Beyond slides]* (tested)

```
import numpy as np
rng = np.random.default_rng(1)
paths, steps, T = 200_000, 500, 2.0
dt = T / steps
dW = np.sqrt(dt) * rng.standard_normal((paths, steps))
W = np.cumsum(dW, axis=1)
W_left = np.hstack([np.zeros((paths, 1)), W[:, :-1]])   # left end point (Ito)
I = np.sum(W_left * dW, axis=1)                          # sum W_{t_{i-1}} dW_i
print(round(I.mean(), 3), round(I.var(), 3), T**2/2)     # -0.0 2.004 2.0
print(round(np.mean(np.abs(I - (0.5*W[:, -1]**2 - 0.5*T))), 3))  # 0.05, shrinks as steps grow
QV = np.sum(dW**2, axis=1)
print(round(QV.mean(), 3), round(QV.std(), 3))           # 2.0 0.127: sum of dW^2 ~ T
```

2 Comparison tables

2.1 Riemann vs stochastic integral

	Riemann $\int f(t)\,\mathrm{d}t$ [S6–S7]	Stochastic $\int f\,\mathrm{d}W_t$ [S8]
Sum	$\sum f(x_i)(t_i - t_{i-1})$	$\sum f(t_{i-1})(W_{t_i} - W_{t_{i-1}})$
Evaluation point	any $x_i \in [t_{i-1}, t_i]$	left endpoint only (previsible)
Result	a number	a random variable
Limit	ordinary	mean square *[Beyond slides]*
Mean / second moment	—	0 / $\mathbb{E}\int f^2\,\mathrm{d}t$ (isometry)

2.2 Ordinary vs Itô calculus

	Ordinary [S19]	Itô [S20–S23]
Second-order term	$(\Delta t)^2 \to 0$	$(\Delta W)^2 \to \mathrm{d}t$ survives
Chain rule	$\mathrm{d}f(x) = f'\,\mathrm{d}x$	$\mathrm{d}f(X) = f'\,\mathrm{d}X + \frac{1}{2}f''(\,\mathrm{d}X)^2$
$\mathrm{d}(W^2)$	$2W\,\mathrm{d}W$	$2W\,\mathrm{d}W + \mathrm{d}t$
$\int_0^T W\,\mathrm{d}W$	$\frac{1}{2}W_T^2$	$\frac{1}{2}W_T^2 - \frac{1}{2}T$
$\mathrm{d}\ln S$ for GBM	$\mu\,\mathrm{d}t + \sigma\,\mathrm{d}W$	$(\mu - \frac{1}{2}\sigma^2)\,\mathrm{d}t + \sigma\,\mathrm{d}W$

2.3 Itô's formula: the two versions

	$Y = f(X_t)$ [S22]	$Y = g(t, X_t)$ [S23]
Drift	$\mu_t f' + \frac{1}{2}\sigma_t^2 f''$	$g_t + \mu_t g_x + \frac{1}{2}\sigma_t^2 g_{xx}$
Volatility	$\sigma_t f'$	$\sigma_t g_x$
Slide example	X^2: $(2\mu X + \sigma^2)\,\mathrm{d}t + 2\sigma X\,\mathrm{d}W$	e^{X+t}: $(1 + \mu + \frac{1}{2}\sigma^2)Y\,\mathrm{d}t + \sigma Y\,\mathrm{d}W$

3 Master formula sheet

Models and SDEs [S2–S5]

- $S_t = S_0 + \mu t + \sigma W_t \sim \mathcal{N}(S_0 + \mu t, \sigma^2 t)$.
- $\mathrm{d}S_t = \mu(t, S_t)\,\mathrm{d}t + \sigma(t, S_t)\,\mathrm{d}W_t \iff S_T = S_0 + \int_0^T \mu\,\mathrm{d}u + \int_0^T \sigma\,\mathrm{d}W_u$.

Stochastic integral [S8–S12]

- $\int_0^T f\,\mathrm{d}W = \lim \sum f(t_{i-1})(W_{t_i} - W_{t_{i-1}})$ (left endpoint).
- $\mathbb{E}[\int f\,\mathrm{d}W] = 0$; $\mathbb{E}[(\int f\,\mathrm{d}W)^2] = \mathbb{E}[\int f^2\,\mathrm{d}t]$; f deterministic $\Rightarrow \mathcal{N}(0, \int f^2\,\mathrm{d}t)$.
- $\int_0^T W\,\mathrm{d}W = \frac{1}{2}W_T^2 - \frac{1}{2}T$: mean 0, variance $T^2/2$.

Box calculus [S13–S14]: $(\mathrm{d}t)^2 = 0,\ \mathrm{d}t\,\mathrm{d}W = 0,\ (\mathrm{d}W)^2 = \mathrm{d}t;\ (\mathrm{d}X)^2 = \sigma^2\,\mathrm{d}t$.

Itô's formula [S20–S23]

- $\mathrm{d}f(W_t) = f'(W_t)\,\mathrm{d}W_t + \frac{1}{2}f''(W_t)\,\mathrm{d}t$.
- $\mathrm{d}f(X_t) = (\mu_t f' + \frac{1}{2}\sigma_t^2 f'')\,\mathrm{d}t + \sigma_t f'\,\mathrm{d}W_t$.
- $\mathrm{d}g(t, X_t) = (g_t + \mu_t g_x + \frac{1}{2}\sigma_t^2 g_{xx})\,\mathrm{d}t + \sigma_t g_x\,\mathrm{d}W_t$.
- GBM: $\mathrm{d}S = \mu S\,\mathrm{d}t + \sigma S\,\mathrm{d}W \Rightarrow S_t = S_0 e^{(\mu - \frac{1}{2}\sigma^2)t + \sigma W_t}$ *[Beyond slides].*

4 Glossary

- **Box calculus:** multiplication table for $\mathrm{d}t$ and $\mathrm{d}W$ [S14].
- **Drift / diffusion coefficient:** $\mu(t, S_t)$ scales $\mathrm{d}t$; $\sigma(t, S_t)$ scales $\mathrm{d}W$ [S3].
- **Itô process:** $\mathrm{d}X = \mu_t\,\mathrm{d}t + \sigma_t\,\mathrm{d}W$ [S22]. **Itô isometry:** $\mathbb{E}[(\int X\,\mathrm{d}W)^2] = \mathbb{E}[\int X^2\,\mathrm{d}t]$ [S9].
- **Itô's formula / lemma:** chain rule for stochastic processes [S22–S23].
- **Previsibility:** integrand known at the start of each interval [S8].
- **Quadratic variation:** $\sum(\Delta W)^2 \to T$ *[Beyond slides].* **SDE:** stochastic differential equation [S5].
- **Stochastic integral:** limit of left-endpoint sums against ΔW [S8].
- **Strong law of large numbers:** sample averages converge to the mean with probability 1 [S13].

5 Self-test

Questions

Q1. ([S2]) For $S_t = S_0 + \mu t + \sigma W_t$ give the distribution of S_t and $\mathbb{P}(S_t < 0)$.

Q2. ([S4–S5]) Why is the drift term "deterministic" over a step, and why can't we write $\mathrm{d}S/\,\mathrm{d}t$?

Q3. ([S8]) Why is the stochastic integral defined with left endpoints?

Q4. ([S9]) Find the distribution of $\int_0^T t\,\mathrm{d}W_t$.

Q5. ([S12]) Find the mean and variance of $\int_0^T W_t\,\mathrm{d}W_t$.

Q6. ([S14]) Compute $(\mathrm{d}X)^2$ for $\mathrm{d}X = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W$ and $(\mathrm{d}S)^2$ for $\mathrm{d}S = \mu S\,\mathrm{d}t + \sigma S\,\mathrm{d}W$.

Q7. ([S22]) $\mathrm{d}X = \mu\,\mathrm{d}t + \sigma\,\mathrm{d}W$, $Y = X^2$. Find $\mathrm{d}Y$.

Q8. ([S23]) Same X, $Y = e^{X+t}$. Find $\mathrm{d}Y$.

Q9. Use Itô's formula to show $\int_0^T W\,\mathrm{d}W = \frac{1}{2}W_T^2 - \frac{1}{2}T$.

Q10. Solve $\mathrm{d}S = \mu S\,\mathrm{d}t + \sigma S\,\mathrm{d}W$ and find $\mathbb{E}[S_t]$.

Solutions

S1. $\mathcal{N}(S_0 + \mu t, \sigma^2 t)$; $\mathbb{P}(S_t < 0) = \Phi\big(-\frac{S_0 + \mu t}{\sigma\sqrt{t}}\big) > 0$.

S2. $\mu(t, S_t)$ uses the price at the start of the step, already known. W is nowhere differentiable; the SDE means the integral equation.

S3. Previsibility: the integrand (a position) is fixed before the increment is revealed; other points give different limits.

S4. Deterministic integrand: $\mathcal{N}(0, \int_0^T t^2\,\mathrm{d}t) = \mathcal{N}(0, T^3/3)$.

S5. Mean 0; variance $\int_0^T \mathbb{E}[W_t^2]\,\mathrm{d}t = T^2/2$.

S6. $\sigma^2\,\mathrm{d}t$; $\sigma^2 S^2\,\mathrm{d}t$.

S7. $\mathrm{d}Y = (2\mu X + \sigma^2)\,\mathrm{d}t + 2\sigma X\,\mathrm{d}W$.

S8. $\mathrm{d}Y = (1 + \mu + \frac{1}{2}\sigma^2)Y\,\mathrm{d}t + \sigma Y\,\mathrm{d}W$.

S9. $f(w) = w^2$: $\mathrm{d}(W^2) = 2W\,\mathrm{d}W + \mathrm{d}t$; integrate: $W_T^2 = 2\int W\,\mathrm{d}W + T$.

S10. $\mathrm{d}\ln S = (\mu - \frac{1}{2}\sigma^2)\,\mathrm{d}t + \sigma\,\mathrm{d}W$, $S_t = S_0 e^{(\mu-\frac{1}{2}\sigma^2)t+\sigma W_t}$; $\mathbb{E}[S_t] = S_0 e^{(\mu-\frac{1}{2}\sigma^2)t} e^{\frac{1}{2}\sigma^2 t} = S_0 e^{\mu t}$.

Appendix A. Slide coverage checklist

Slide	Content	Section
S2	$S_t = S_0 + \mu t + \sigma W_t$; mean and variance	1.1
S3	Integral and differential forms; state-dependent coefficients	1.2
S4–S5	Deterministic and stochastic components; $\mathrm{d}S/\,\mathrm{d}t$ not feasible	1.3
S6–S7	Integral form; Riemann integral	1.4
S8	Stochastic integral; previsibility	1.5
S9	Properties; Itô isometry	1.6
S10–S11	Proofs of (1) and (2)	1.7
S12	Example $\int W\,\mathrm{d}W$	1.8
S13–S14	Strong law; box calculus	1.9
S15–S18	History; why Itô's formula	1.10
S19–S21	Taylor: ordinary, $f(W_t)$, $f(S_t)$	1.11
S22	Itô's formula; example X^2	1.12
S23	Itô's formula with time; example e^{X+t}	1.13

Appendix B. Flag summary (△!)

1. [S3] "A stochastic process is a continuous process that can be written …": this defines an Itô process, not every stochastic process.
2. [S9] Property (3) needs a deterministic integrand; (2) and (5) are the same statement.
3. [S10–S11] Exchange of $\mathbb{E}$ and lim is justified by mean-square convergence, not shown.
4. [S13] The strong law should read $\mathbb{P}(\lim_n \frac{1}{n}\sum X_i = \mu) = 1$. The real reason $(\Delta W)^2 \approx \Delta t$ is quadratic variation $\sum(\Delta W)^2 \to T$.
5. [S4, S12, S22, S23] Questions and examples posed without worked answers; the solutions here are my own.